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Maternal pre-pregnancy overweight and infertility in sons and daughters: A cohort study

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Abstract

Introduction: Overweight and obesity in pregnancy is increasing worldwide and may harm the developing fetus, including its future reproductive health. We therefore studied the association between in utero exposure to maternal overweight and obesity and infertility in adulthood. No studies have previously assessed this association.

Material and methods: We performed a cohort study with 9232 adult sons and daughters whose mothers were enrolled in the Danish Healthy Habits for Two cohort during pregnancy in 1984-87. Participants were sons and daughters followed in the Danish In-Vitro-Fertilization-Register and Danish National Patient Register until February 2018 for diagnoses of infertility.

Results: In total, 1203 (13%) sons and daughters were born to mothers with a body mass index (BMI) >25 kg/m^2; 871 (9.4%) of the participants were identified as being infertile during follow-up. Sons of overweight mothers had slightly increased odds of infertility compared with sons of mothers with normal body weight (BMI 18.5-24.9 kg/m^2, adjusted odds ratio 1.4, 95% confidence interval [CI] 1.0-1.9). Cubic spline analyses with continuous BMI levels showed increasing odds with higher levels of BMI; however, for BMI >29 kg/m^2 the confidence intervals were too wide to draw conclusions. No association between maternal overweight and infertility was found among daughters (adjusted odds ratio 0.9, 95% CI 0.7-1.2)).

Conclusions: Sons born to overweight mothers had higher odds of infertility compared with sons of normal weight mothers. No association between maternal overweight and infertility was observed in daughters. Prevention of overweight during pregnancy may be an important tool to preserve fecundity in future generations.

KEYWORDS

cohort study, infertility, male offspring, maternal overweight, pregnancy, prenatal exposure delayed effect, reproductive health

Abbreviations: BMI, body mass index; CI, confidence interval; DNPR, Danish National Patient Register; ICD-10, International Classification of Diseases, 10th revision; IVF, in vitro fertilization; OR, odds ratio.

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1 | INTRODUCTION

During recent decades, fertility rates have declined in many high income countries.[1] The tendency of delayed childbearing has a substantial influence on fecundity, but the increasing number of young couples seeking treatment for infertility suggests that other factors also play a role.[2]

A variety of prenatal factors, such as maternal tobacco use, exposure to endocrine-disrupting chemicals and poor nutrition, may induce long-lasting alterations in the reproductive system of the fetus.[3] Further, the prevalence of overweight and obesity has increased since the 1980s.[4] In adulthood, overweight is associated with reduced fecundity in both men and women.[5-7] Maternal overweight during pregnancy is furthermore known to have long-term health consequences for the child,[8] but research has mostly focused on complications in the metabolic and cardiovascular system of offspring.[8] Overall, the impact of maternal overweight and obesity on reproductive health of the child is not well understood.

Maternal overweight could affect the offspring's reproductive health through several potential mechanisms. Fatty tissue is hormonally active, and fetal exposure to, for example, leptin, androgens and estrogens has been suggested to interfere with development of the reproductive system and induce changes which might not manifest until sexual maturity.[9,10] Some endocrine-disrupting chemicals accumulate in adipose tissue and may become bioavailable and enter the maternal and the fetal bloodstream during remodeling of maternal fat stores during pregnancy.[11,12] Obesity is also associated with low-grade metabolic inflammation,[13] a proposed factor in programming of reproductive fitness early in life.[3]

Studies on daughters suggest that maternal overweight might be associated with earlier age of menarche and attained breast stages as well as other hormonal markers of reproductive health.[14-19] Studies on sons have indicated that maternal overweight is associated with anomalies in the male genital system[20] and a tendency to earlier timing of pubertal milestones and reduced semen quality.[21,22]

To the best of our knowledge, no previous studies have investigated the association between maternal overweight during pregnancy and infertility in their sons or daughters. The aim of this study was to investigate the association between pre-pregnancy maternal body mass index (BMI) and infertility in adult sons and daughters.

2 | MATERIAL AND METHODS

2.1 | Study population

The study population included sons and daughters whose mothers were recruited to the Healthy Habits for Two cohort.[23] This cohort was established as a health campaign enrolling all pregnant women from April 1984 to April 1987 in the Danish municipalities of Odense and Aalborg. The women were enrolled during the 36th gestational week at a scheduled visit at the midwifery clinic and were asked to fill out a questionnaire regarding health, lifestyle and sociodemographic factors. Of a total of 11 980 pregnant women (87% of the eligible), 11 144 delivered live born singletons.

Key message

In this Danish cohort study, sons born to overweight mothers had a higher risk of infertility in adulthood compared with sons of normal weight mothers. No association was found among daughters.

At the time of follow-up in February 2018, the study population of sons and daughters was between 31 and 34 years old. Information on sons and daughters who had died or emigrated were available from the Danish Civil Registration System.[24] Sons and daughters were excluded if information on the mothers' pre-pregnancy height or weight were missing, if they were not alive or if they had lived outside Denmark for at least 6 months after their 18th birthday. A flowchart of the establishment of the study population is provided in Figure 1.

2.2 | Assessment of maternal pre-pregnancy BMI

When the mothers from the Healthy Habits for Two cohort attended the first antenatal visit between gestational weeks 6 and 9, they reported their height and pre-pregnancy weight, and their general practitioner measured their current weight. These data were later extracted from the mothers' medical records together with birth data.

Maternal BMI (kg/m^2) was calculated by dividing the pre-pregnancy weight in kilograms by the height in meters squared. BMI was classified according to the World Health Organization

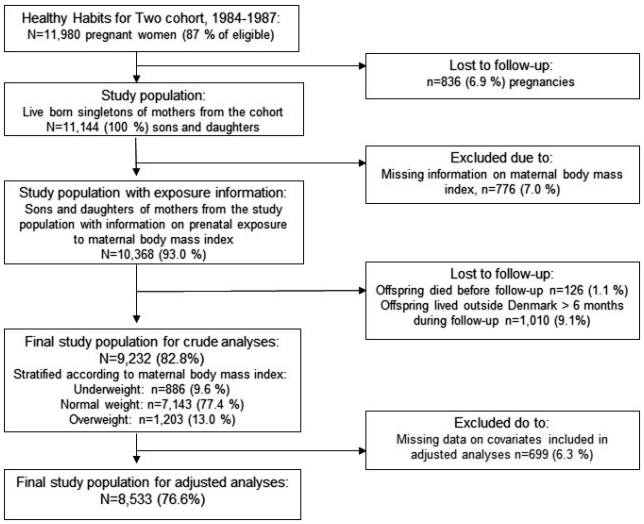

FIGURE 1 Flowchart of the study population from Healthy Habits for Two (1984-87) followed up in registers in 2018

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criteria; as underweight (BMI <18.5 kg/m^2), normal weight (BMI 18.5-24.9 kg/m^2), overweight (BMI 25-29.9 kg/m^2) and obese (BMI ≥30 kg/m^2). As expected from the low prevalence of obesity in the 1980s, the number of obese mothers in the study population was low. Unfortunately, the number was too low to report the association separately for overweight and obese mothers, and thus maternal overweight and obesity were combined into one group referred to as "maternal overweight" (BMI ≥25 kg/m^2). In the subanalysis described below, we also used maternal BMI as a continuous variable.

2.3 | Information on infertility

Two national health registers provided information on infertility in sons and daughters during follow-up: The Danish National Patient Register (DNPR)[25] and the In Vitro Fertilization (IVF) register.[26]

DNPR contains information on all inpatient contacts to Danish hospitals since 1977 and outpatient contacts from 1995. In the DNPR, diagnoses are classified according to the International Classification of Diseases, 10th revision (ICD-10), since 1994. Daughters were considered infertile if they had one of the following diagnoses; N97 (female infertility), N97.0 (female infertility associated with anovulation), N97.1 (female infertility of tubal origin), N97.2 (female infertility of uterine origin), N97.3 (female infertility of cervical origin), N97.8 (female infertility of other origin), N97.9 (female infertility of unspecified origin). Sons were considered infertile if they were diagnosed with N46 (male factor infertility).

The Danish IVF register was established in 1994 and contains information on all treatments with in vitro fertilization, intracytoplasmic sperm injection, frozen embryo replacements and egg donations carried out at public and private fertility clinics in Denmark. The register contains clinical as well as laboratory data, such as the primary and secondary cause of treatment, which are registered with ICD-10 codes. The IVF register holds information on both individuals if they are in a relationship.

Sons and daughters were defined as infertile if they were present in one or both health registers with the aforementioned diagnoses and/or treatments. This definition did not account for the fact that some individuals might be present in the IVF register because their partner was subfertile. To increase the precision of this classification, we further restricted the definition of infertility in two subanalyses (models 2 and 3). In model 2, individuals registered in the IVF register were excluded from the definition as follows: egg donors; sons whose partners were only registered as egg donors; daughters registered as "living alone" (ICD-10 code: Z60.2) without a registered infertility diagnosis (treatment because of no male partner); daughters registered with "female infertility associated with male factor" (ICD-10 code: N97.4); and sons registered with "normal semen quality" (Code: EZDH101) whose partner had an infertility diagnosis. In model 3, individuals with an unknown cause of infertility treatment were excluded.

2.4 | Covariates

The covariates included in the adjusted analysis were chosen a priori based on knowledge on the subject matter from published literature and directed acyclic graphs (DAGs), which are causal diagrams used to make decisions on analytic strategy. The information was available from the questionnaires and the health registers and included: maternal age in years (continuous), maternal parity (0 or ≥1), parents' cohabitation status (living together, yes or no), maternal cigarette smoking during pregnancy (no, yes but stopped <36th gestational week, yes 0-9 or yes ≥10 cigarettes/d), maternal alcohol consumption during pregnancy (<1, 1-4, >4 drinks/wk), and socioeconomic status of the parents based on the highest ranking of the mother's and her partner's occupational status at the time of pregnancy (high, middle, low). The latter was based on several questions in the Healthy Habits for Two questionnaire, which resulted in an occupational code according to the Danish version of the International Standard Classification of Occupations (DISCO-08). The codes were additionally grouped into the three categories, which were equivalent to educational level.

2.5 | Statistical analyses

The characteristics of the study population are presented as means and standard deviations for the continuous variables and as number and percentages for categorical variables, stratified by maternal pre-pregnancy BMI. Numbers and percentages of infertile individuals identified in the two registers and included in each of the three models of analysis were calculated. These are presented for the total study population as well as for sons and daughters separately and according to each of the four birth years in the cohort (1984-87).

The association between maternal BMI and infertility was assessed using logistic regression analyses estimating crude and adjusted odds ratios (ORs) with 95% confidence intervals (CIs) based on the different definitions of infertility. Maternal BMI in the three groups constituted the independent variable with normal weight being the reference. The binary outcome was infertility. We performed three models, with different outcome classification, as described above. All analyses were stratified by gender.

In a subanalysis, we performed a cubic spline analysis with continuous BMI.

All analyses were performed using STATA version 15 software (Stata Corp.). Since the cohort contained some siblings, all analyses were performed allowing for clustering under the mothers' identification number, and the results presented are calculated based on robust standard errors.

2.6 | Ethical approval

The data collection was approved by the regional ethics committee when the Health Habits for Two Cohort was initiated (1984) and the medical departments involved accepted the data collection.

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TABLE 1 Characteristics of the study population of sons and daughters from the Healthy Habits for Two cohort (1984-87), stratified by their prenatal exposure to maternal body mass index

	Maternal underweight (BMI: <18.5 kg/m^2) n = 886	Maternal normal weight (BMI: 18.5-24.9 kg/m^2) n = 7143	Maternal overweight (BMI: ≥25 kg/m^2) n = 1203	Total (n = 9232)
Sons, n (%)	421 (47.5)	3699 (51.8)	610 (50.7)	4730 (51.2)
Daughters, n (%)	465 (52.5)	3444 (48.2)	593 (49.3)	4502 (48.8)
Pregnancy-related characteristics				
Mothers maternal age, mean (SD)	26.3 (4.3)	27.2 (4.6)	27.3 (5.0)	27.2 (4.6)
Mother's parity, n (%)				
Nulliparous	437 (49.3)	3537 (49.5)	486 (40.4)	4460 (48.3)
Multiparous	449 (50.7)	3606 (50.5)	717 (59.6)	4772 (51.7)
Maternal smoking, n (%)				
No	376 (42.4)	3956 (55.4)	716 (59.5)	5048 (54.7)
Yes, but mother stopped <36th GW	47 (5.3)	382 (5.4)	56 (4.7)	485 (5.3)
Yes, 0-19 cigarettes/d	211 (23.8)	1318 (18.5)	170 (14.1)	1699 (18.4)
Yes, ≥10 cigarettes/d	241 (27.2)	1418 (19.9)	247 (20.5)	1906 (20.7)
Maternal alcohol consumption, n (%)				
<1 drink/wk	368 (41.5)	2375 (33.3)	500 (41.6)	3243 (35.1)
2-4 drinks/wk	452 (51.0)	4184 (58.6)	634 (52.7)	5270 (57.1)
>4 drinks/wk	65 (7.3)	580 (8.1)	69 (5.7)	714 (7.7)
Parents' highest occupational level, n (%)				
Low	237 (26.8)	1882 (26.4)	422 (35.1)	2541 (27.5)
Middle	477 (53.8)	3902 (54.6)	613 (51.0)	4992 (54.1)
High	167 (18.9)	1304 (18.3)	157 (13.1)	1628 (17.6)
Parents' cohabitation status, n (%)				
Living together	800 (90.3)	6521 (91.3)	1105 (91.9)	8426 (91.3)
Not living together	29 (3.3)	191 (2.7)	43 (3.6)	263 (2.9)
Missing	57 (6.4)	431 (6.0)	55 (4.6)	543 (5.9)

According to the data confidentiality policy of Statistics Denmark, no fewer than five cases per cell must be reported or inferred, here indicated by a -. If missing is not stated, it is between 0 and 1

Abbreviations: BMI, body mass index; GW, gestational week; SD, standard deviation.

All participants recruited in the Health Habits for Two Cohort provided written informed consent upon recruitment. The Danish Data Protection Agency approved the storage and use of the birth cohort data on 22 July 1993 (1993-1110-1305) and the present study was registered at Aarhus University's Data Inventory (2016-051-000001-749).

3 | RESULTS

The final study population consisted of 9232 (83%) children, 4730 (51%) sons and 4502 (49%) daughters (Figure 1). Of these children, 886 (10%) were born to underweight mothers, 7143 (77%) to normal weight mothers and 1203 (13.0%) to overweight mothers. In the adjusted analyses, a further 699 (8%) sons and daughters were excluded due to missing information on covariates.

Table 1 presents the characteristics of the study population stratified by maternal BMI. A higher proportion of sons and daughters exposed to maternal overweight had mothers who were multiparous, did not smoke, drank <1 drink/wk, and parents at the lowest occupational level, compared with sons and daughters of mothers with normal weight. Characteristics and descriptive results on the excluded children are presented in Tables S1 and S2. The prevalence of infertility was similar to the final study population.

The total numbers of infertile individuals are presented in Table 2. A total of 569 (6%) individuals were identified with an infertility diagnosis in the DNPR, and 670 (7%) were identified as being treated for infertility by the IVF register.

The main results are provided in Table 3. Sons exposed to maternal overweight had consistently higher ORs for infertility compared with sons of normal weight mothers (model 1: OR 1.4, 95% CI 1.0-1.9). Sons exposed to maternal underweight did not present with increased risk of infertility compared with the reference group (OR 1.1, 95% CI 0.7-1.6). Neither maternal overweight (OR 0.9, 95% CI 0.7-1.2) nor maternal underweight (OR 1.0, 95% CI 0.8-1.4) were associated with infertility in daughters. All results were approximately similar in the three models.

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TABLE 2 Numbers of infertile sons and daughters from the Healthy Habits for Two cohort (1984-87), identified in the two Danish national administrative registers; the Danish National Patient Register and the In Vitro Fertilization register

	Infertile individuals identified in the two registers		Infertile individuals in the three models		
	DNPR, n (%)	IVF register, n (%)	Model 1, n (%)	Model 2, n (%)	Model 3, n (%)
Total study population (n = 9232)	569 (6.2)	670 (7.3)	871 (9.4)	766 (8.3)	688 (7.5)
Sons (n = 4730)	204 (4.3)	270 (5.7)	354 (7.5)	299 (6.3)	254 (5.4)
Daughters (n = 4502)	365 (8.1)	400 (8.9)	517 (11.5)	467 (10.4)	434 (9.6)
Birth cohorts					
1984 (n = 2103)	156 (7.4)	183 (8.7)	225 (10.7)	198 (9.4)	185 (8.8)
1985 (n = 3216)	204 (6.3)	235 (7.3)	313 (9.7)	278 (8.6)	245 (7.6)
1986 (n = 3156)	175 (5.5)	206 (6.5)	276 (8.7)	241 (7.6)	216 (6.8)
1987 (n = 757)	34 (4.5)	46 (6.1)	57 (7.5)	49 (6.5)	42 (5.5)

Model 1 included all sons and daughters who had an infertility diagnosis in the DNPR and/or who were present in the IVF register.

Model 2 excluded daughters and sons who most were likely fertile.

Model 3 included only individuals who most likely were infertile.

Abbreviations: DNPR, Danish national patient register; IVF, In vitro fertilization register.

TABLE 3 Crude and adjusted odds ratio for infertility among sons and daughters from the Healthy Habits for Two cohort (1984-87) in relation to maternal body mass index stratified by sex

	Sons			Daughters		
	Maternal underweight (BMI <18.5 kg/m^2) n = 421	Maternal normal weight (BMI = 18.5-24.9 kg/m^2) n = 3699	Maternal overweight (BMI ≥25 kg/m^2) n = 610	Maternal underweight (BMI <18.5 kg/m^2) n = 465	Maternal normal weight (BMI = 18.5-24.9 kg/m^2) n = 3444	Maternal overweight (BMI ≥25 kg/m^2) n = 593
Main analysis						
Model 1						
No. of infertile individuals (%)	32 (7.6)	260 (7.0)	62 (10.2)	59 (12.7)	392 (11.4)	66 (11.1)
cOR (95% CI)	1.1 (0.7-1.6)	1.0 (reference)	1.5 (1.1-2.0)	1.1 (0.8-1.5)	1.0 (reference)	1.0 (0.7-1.3)
aOR (95% CI)	1.1 (0.7-1.6)	1.0 (reference)	1.4 (1.0-1.9)	1.0 (0.8-1.4)	1.0 (reference)	0.9 (0.7-1.2)
Subanalyses						
Model 2						
No. of infertile individuals (%)	28 (6.7)	220 (6.0)	52 (8.4)	54 (11.6)	355 (10.3)	58 (9.8)
cOR	1.1 (0.8-1.7)	1.0 (reference)	1.4 (1.1-2.0)	1.1 (0.8-1.5)	1.0 (reference)	0.9 (0.7-1.3)
aOR (95% CI)	1.1 (0.7-1.7)	1.0 (reference)	1.3 (1.0-1.9)	1.0 (0.8-1.4)	1.0 (reference)	0.9 (0.6-1.2)
Model 3						
No. of infertile individuals (%)	21 (5.0)	186 (5.0)	47 (7.7)	52 (11.2)	330 (9.6)	52 (8.8)
cOR	1.0 (0.6-1.6)	1.0 (reference)	1.6 (1.1-2.2)	1.2 (0.9-1.6)	1.0 (reference)	0.9 (0.7-1.2)
aOR (95% CI)	0.9 (0.6-1.5)	1.0 (reference)	1.5 (1.0-2.1)	1.1 (0.8-1.5)	1.0 (reference)	0.8 (0.6-1.1)

Adjusted for maternal age (years), parity (0 and ≥1), maternal smoking in pregnancy (no, yes but stopped <36 weeks of gestation, 0-10 cigarettes/d or >10 cigarettes/d), maternal alcohol consumption (<1, 2-4 or >4.5 drinks/weekly), highest parental occupational level (low, middle or high) and parental cohabitation (living together or not living together).

Model 1 included all sons and daughters who had an infertility diagnose in the DNPR and/or who were present in the IVF register.

Model 2 excluded daughters and sons who most likely were fertile.

Model 3 included only individuals who most likely were infertile.

Abbreviations: aOR, adjusted odds ratio; BMI, body mass index; CI, confidence interval; cOR, crude odds ratio.

The results from the cubic spline analysis are presented in Figure 2A,B. For adult boys (Figure 2A), the results interestingly show a U-shaped curve with progressively increasing odds with higher levels of BMI; however, for BMI >29 kg/m^2, the confidence intervals were too wide to draw any conclusions. For girls (Figure 2B), there was no association with maternal overweight.

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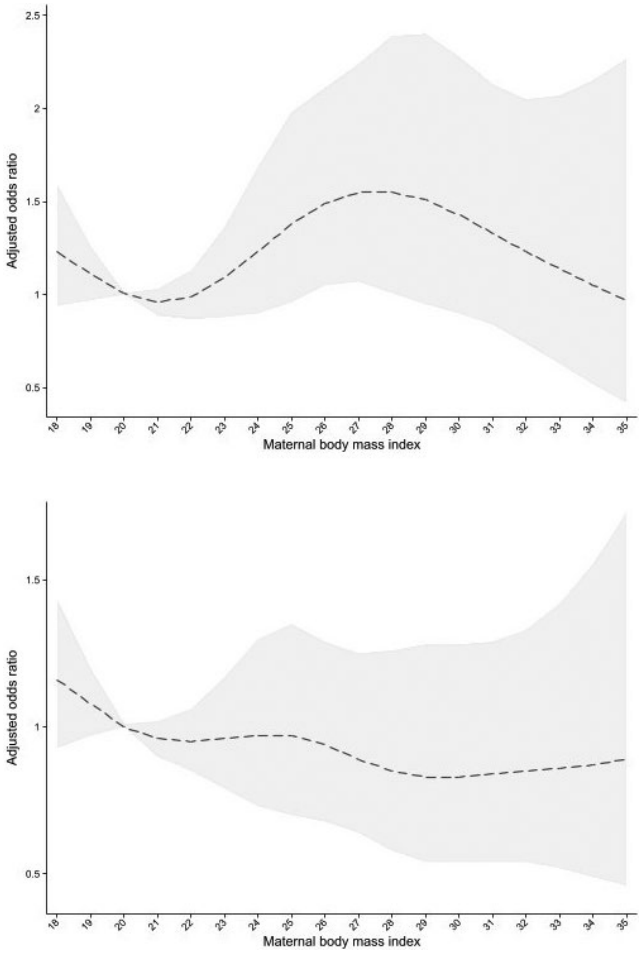

FIGURE 2 A, Association between maternal BMI and infertility among sons from the Healthy Habits for Two cohort (1984-87). Cubic spline analysis. B, Association between maternal BMI and infertility among daughters from the Healthy Habits for Two cohort (1984-87). Cubic spline analysis

4 | DISCUSSION

To the best of our knowledge, this is the first study to estimate the association between maternal overweight during pregnancy and infertility in adulthood. We found that sons born to overweight mothers had higher odds of infertility compared with sons of normal weight mothers. No association between maternal overweight and infertility was observed in daughters.

Only a few previous studies have investigated associations between maternal overweight and aspects of reproductive health in children. In a pilot study of sons from this cohort, Ramlau-Hansen et al indicated an association between higher maternal BMI and lower semen quality in sons, although the study only had power to provide preliminary evidence.[22] In another study from this cohort, Hounsgaard et al found that maternal obesity might be related to earlier timing of pubertal milestones in sons.[21] Two recent studies have also shown that maternal pre-pregnancy obesity is associated with earlier pubertal milestones, albeit partly mediated through childhood overweight.[27] Finally, a large Swedish register-based study recently reported that maternal overweight and obesity were associated with increased occurrence of hypospadias and cryptorchidism in the sons. Interestingly, this study also revealed that the association was partly mediated through obesity-related diseases, mainly preeclampsia.[20]

Most studies on the association between maternal overweight and reproductive health in daughters used age at menarche as the primary outcome and found that maternal overweight and/or obesity were associated with earlier age at menarche.[14-16,18,19]

The potential mechanisms underlying the association between maternal overweight and offspring reproductive health are not known but several hypotheses have been proposed. Maternal overweight is an indicator of a changed hormonal balance, which could interfere with development of the offspring's reproductive system,[9,10] possibly via low-grade maternal inflammation induced by obesity[3] but other possible pathways might link maternal overweight and reproductive health in the offspring. It has been suggested, that the association between maternal overweight and daughter's age at menarche might be due to a shared environment during upbringing or to genetic variants, and that the association might be mediated by the offspring's own pubertal BMI.[18,19]

Among the strengths of this study was collection of data on maternal pre-pregnancy BMI during early pregnancy. This reduces the risk of differential recall bias relative to the exposure. Another strength was the high participation rate at enrollment of the pregnant women (87%) and the unique opportunity to link data from the cohort with outcome data from easily accessible and reliable national administrative registers. We excluded sons and daughters with missing information on maternal pre-pregnancy BMI, loss to follow up and missing data on covariates, which poses a risk of selection bias due to missing data. Characteristics of the excluded sons and daughters differed across the three exposure groups, but the prevalence of infertility was similar to the final study population, except in relation to those excluded due to loss to follow up, as expected. Non-response might be related to lower sociodemographic status[28] and BMI has a strong inverse association with sociodemographic status among women.[4] If this applies to the present study, and if sociodemographic status likewise is associated with infertility, it might result in bias, most likely causing an underestimation of the observed association.

The prevalence of infertility in this study population was lower than reported in population surveys, in which about 15% of all couples experience infertility. This is most likely a consequence of the follow up at a relatively young age, where the sons and daughter had not all reached the end of their reproductive lives. In addition, only individuals trying to become pregnant and seek help and are present in the DNPR and IVF register, and help-seekers tend to be of higher social class.[29] Children's and parents' social class are associated,[30] and overweight is likewise associated with social class.[31] If this applies to the present study, fewer sons and daughters exposed to maternal overweight may seek medical help and thereby become registered as infertile. Such differential misclassification could lead to bias, most likely causing underestimation of the association.

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5 | CONCLUSION

This study supports the hypothesis that maternal overweight affects reproductive health in male offspring, but further studies are needed.

CONFLICT OF INTEREST

None.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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